

## Literature review evidence table — sources R01-R02 (1/11)

| Reference                                                                                                                                    | Main contribution / evidence                                             | Role in thesis                                                                    | Source and citation status                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------|
| R01   Workload foundations<br>Vaswani et al. (2017). Attention Is All You Need. NeurIPS.                                                     | Defines the Transformer and scaled dot-product self-attention.           | Provides the attention workload foundation; report prefill and decode separately. | Peer-reviewed. Found in DOCX and PPTX.                               |
| R02   Workload foundations<br>Devlin et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL. | Shows the Transformer as a reusable large-scale language-model workload. | Justifies realistic Transformer inference evaluation and model selection.         | Peer-reviewed. PPTX only; citation completed from standard metadata. |

Yellow rows need citation completion or verification before thesis submission.

Literature review evidence table — sources R03-R04 (2/11)

| Reference                                                                                                                                                          | Main contribution / evidence                                                     | Role in thesis                                                                       | Source and citation status    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------|
| R03   Workload foundations<br>Sze et al. (2017). Efficient Processing of Deep Neural Networks: A Tutorial and Survey. Proceedings of the IEEE, 105(12), 2295–2329. | Surveys data reuse, energy, throughput, hardware, and evaluation trade-offs.     | Supports measuring memory, latency, and throughput together rather than FLOPs alone. | Peer-reviewed. DOCX and PPTX. |
| R04   Platforms and profiling<br>Shen et al. (2018). CUDAAAdvisor: LLVM-Based Runtime Profiling for Modern GPUs. CGO, 214–227.                                     | Profiles runtime and memory behavior across CUDA versions and GPU architectures. | Use profiler evidence to attribute a speedup; do not rely on elapsed time alone.     | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R05-R06 (3/11)

| Reference                                                                                                                                                           | Main contribution / evidence                                                        | Role in thesis                                                                             | Source and citation status    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------|
| R05   Platforms and profiling<br>Mittal & Vaishay (2019). A Survey of Techniques for Optimizing Deep Learning on GPUs. Journal of Systems Architecture, 99, 101635. | Classifies architecture- and system-level GPU optimization methods.                 | Provides a CUDA taxonomy for interpreting memory, precision, computation, and parallelism. | Peer-reviewed. DOCX and PPTX. |
| R06   Attention and memory efficiency<br>Dao et al. (2022). FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. NeurIPS.                   | Uses tiling and fusion to reduce HBM↔SRAM traffic while preserving exact attention. | Primary portable principle: reduce high-bandwidth-memory reads and writes.                 | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R07-R08 (4/11)

| Reference                                                                                                                                                       | Main contribution / evidence                                                               | Role in thesis                                                                       | Source and citation status    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------|
| R07   Attention and memory efficiency<br>Dao (2024). FlashAttention-2:<br>Faster Attention with Better<br>Parallelism and Work Partitioning.<br>ICLR.           | Improves parallel work partitioning<br>beyond the first FlashAttention<br>kernel.          | Compare the IO-aware principle with<br>backend-specific execution mapping.           | Peer-reviewed. DOCX and PPTX. |
| R08   Attention and memory efficiency<br>Shah et al. (2024).<br>FlashAttention-3: Fast and<br>Accurate Attention with Asynchrony<br>and Low-precision. NeurIPS. | Uses Hopper-specific asynchronous<br>pipelines, warp specialization, and<br>low precision. | Key portability-limit example: later<br>gains rely on NVIDIA-specific<br>facilities. | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R09-R10 (5/11)

| Reference                                                                                                                                            | Main contribution / evidence                                                                          | Role in thesis                                                                | Source and citation status    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------|
| R09   Attention and memory efficiency<br>Yuan et al. (2025). Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. ACL. | Combines hierarchical token compression, selection, and local windows with hardware-aligned sparsity. | For long context, measure quality and phase-specific latency, not only FLOPs. | Peer-reviewed. DOCX and PPTX. |
| R10   Attention and memory efficiency<br>Alizadeh et al. (2024). LLM in a Flash: Efficient Large Language Model Inference with Limited Memory. ACL.  | Selectively loads model parameters from flash storage into memory on demand.                          | Keep parameter-residency methods distinct from attention-kernel methods.      | Peer-reviewed. DOCX and PPTX. |

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Literature review evidence table — sources R11-R12 (6/11)

| Reference                                                                                                                                                       | Main contribution / evidence                                                    | Role in thesis                                                                                    | Source and citation status                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| R11   Serving<br>Kwon et al. (2023). Efficient Memory Management for Large Language Model Serving with PagedAttention. SOSP.                                    | Pages the KV cache to reduce fragmentation and improve sharing and batching.    | Measure isolated kernels and end-to-end serving separately; report cache and batching conditions. | Peer-reviewed. DOCX and PPTX.                                          |
| R12   Platforms and profiling<br>Benazir & Lin (2025). Profiling Large Language Model Inference on Apple Silicon: A Quantization Perspective. arXiv:2508.08531. | Profiles Apple Silicon and NVIDIA across model sizes and quantization settings. | Use system-level Metal profiling and state the precision and execution path explicitly.           | Preprint. DOCX and PPTX; local DOCX venue claim requires verification. |

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Literature review evidence table — sources R13-R14 (7/11)

| Reference                                                                                                                                               | Main contribution / evidence                                                                                   | Role in thesis                                                                             | Source and citation status                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| R13   Platforms and profiling<br>Kirk & Hwu (2022). Programming<br>Massively Parallel Processors: A<br>Hands-on Approach (4th ed.).<br>Morgan Kaufmann. | Explains coalescing, tiling,<br>synchronization, occupancy, and CUDA<br>execution.                             | Use as background for CUDA diagnosis;<br>not direct evidence of Metal<br>portability.      | Textbook. PPTX only; citation<br>completed from standard<br>metadata.                  |
| R14   Platforms and profiling<br>Apple ML Research (2025). MLX /<br>Apple Silicon platform evidence<br>(exact report unspecified in local<br>files).    | Materials describe unified memory,<br>Metal Performance Primitives, and<br>distinct prompt/generation regimes. | Use only as contextual platform<br>evidence; keep it distinct from<br>independent studies. | Incomplete. DOCX and PPTX; add<br>exact report, author, date, URL,<br>and access date. |

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## Literature review evidence table — sources R15-R16 (8/11)

| Reference                                                                                                                       | Main contribution / evidence                                                      | Role in thesis                                                                                       | Source and citation status                                                        |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| R15   Serving and deployment<br>Wu et al. (2019). Edge-inference deployment study (title and venue unspecified in local files). | Cited for operator support and hardware fragmentation in edge deployment.         | Document runtime/backend coverage; distinguish Metal GPU, Core ML, and ANE paths.                    | Incomplete. DOCX and PPTX; locate original title, venue, and identifier.          |
| R16   Serving and evaluation<br>MLCommons / MLPerf Inference benchmark suite (exact version unspecified in local files).        | Defines scenario-, quality-, version-, and run-rule-based performance evaluation. | Declare model, precision, shapes, versions, warm-up, repetitions, scenario, and quality constraints. | Incomplete. DOCX and PPTX; cite the precise MLPerf Inference version and ruleset. |

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Literature review evidence table — sources R17-R18 (9/11)

| Reference                                                                                                                                                                                                                                                                                                                    | Main contribution / evidence                                                                                         | Role in thesis                                                                                                                                                        | Source and citation status                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| R17   Serving and evaluation Schryen (2024). Speedup and Efficiency of Computational Parallelization: A Unifying Approach and Asymptotic Analysis. JPDC, 185, 104801.                                                                                                                                                        | Unifies speedup and efficiency models and analyzes overhead and scalability.                                         | Report within-platform relative speedups and explain residual serial and system overhead.                                                                             | Peer-reviewed. DOCX and PPTX.                                |
| R18   Metal and cross-platform execution Gebraad & Fichtner (2023). Seamless GPU Acceleration for C++-Based Physics with the Metal Shading Language on Apple’s M Series Unified Chips. Seismological Research Letters, 94(3), 1670-1675. <a href="https://doi.org/10.1785/0220220241">https://doi.org/10.1785/0220220241</a> | Implements Metal compute kernels on Apple M-series unified-memory systems and evaluates array and stencil workloads. | Use as direct published evidence that Metal’s unified-memory model changes host-device data-management assumptions; validate the same effect for the thesis workload. | Peer-reviewed journal article. Added source; Metal-specific. |

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## Literature review evidence table — sources R19-R20 (10/11)

| Reference                                                                                                                                                                                                                                                                                                                                         | Main contribution / evidence                                                                                                            | Role in thesis                                                                                                                          | Source and citation status                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| R19   Metal and cross-platform execution<br>Black et al. (2025). Accelerating Euclidean Distance Transforms: A Fast and Flexible Approach With Multi-Vendor GPU, Multi-Threading, and Multi-Language Support. IEEE Access, 13, 44636–44649. <a href="https://doi.org/10.1109/ACCESS.2025.3548563">https://doi.org/10.1109/ACCESS.2025.3548563</a> | Demonstrates one implementation across NVIDIA CUDA, AMD ROCm, Apple Metal, and Intel oneAPI, while reporting cross-platform benchmarks. | Supports separating portable algorithm design from backend-specific tuning and reporting the backend and hardware for every comparison. | Peer-reviewed IEEE journal article. Added source; CUDA/Metal cross-platform.   |
| R20   CUDA performance modeling<br>Lym et al. (2019). DeLTA: GPU Performance Model for Deep Learning Applications with In-Depth Memory System Traffic Analysis. ISPASS, 129–139. <a href="https://doi.org/10.1109/ISPASS.2019.00041">https://doi.org/10.1109/ISPASS.2019.00041</a>                                                                | Models deep-learning GPU performance from traffic at multiple memory-hierarchy levels and accounts for reuse in parallel convolution.   | Motivates collecting memory-traffic and reuse evidence alongside elapsed time when diagnosing CUDA attention or inference kernels.      | Peer-reviewed IEEE conference article. Added source; CUDA/GPU memory analysis. |

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Literature review evidence table — sources R21-R22 (11/11)

| Reference                                                                                                                                                                                                                                                                                          | Main contribution / evidence                                                                                                   | Role in thesis                                                                                                                                  | Source and citation status                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| R21   CUDA inference kernels<br>Sun et al. (2022). Accelerating Sparse Deep Neural Network Inference Using GPU Tensor Cores. HPEC. <a href="https://doi.org/10.1109/HPEC.55821.2022.9926300">https://doi.org/10.1109/HPEC.55821.2022.9926300</a>                                                   | Maps locally dense regions to NVIDIA Tensor Cores and sparse regions to CUDA Cores to improve sparse DNN inference throughput. | Shows why an optimization’s benefit may depend on NVIDIA-specific execution units; distinguish portable sparsity ideas from CUDA-only mappings. | Peer-reviewed IEEE conference article. Added source; CUDA/Tensor Cores. |
| R22   CUDA serving and batching<br>Zhang et al. (2024). BatOpt: Optimizing GPU-Based Deep Learning Inference Using Dynamic Batch Processing. IEEE Transactions on Cloud Computing, 12(1), 174–185. <a href="https://doi.org/10.1109/TCC.2024.3350561">https://doi.org/10.1109/TCC.2024.3350561</a> | Uses queueing-based dynamic batching to balance service latency and GPU memory use for DNN inference.                          | Supports reporting batch policy, queueing conditions, memory use, and latency together in end-to-end CUDA/Metal serving experiments.            | Peer-reviewed IEEE journal article. Added source; GPU serving.          |

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